Universiti Tun Hussein Onn Malaysia

UNIVERSITI TUN HUSSEIN ONN MALAYSIA

FINAL EXAMINATION
SEMESTER I
SESSION 2024/2025

COURSE NAME	:	OPERATING SYSTEM
COURSE CODE	:	BIC 20803
PROGRAMME CODE	:	BIS / BIP / BIW / BIM
EXAMINATION DATE	:	FEBRUARY 2025
DURATION	:	3 HOURS
INSTRUCTIONS	:	1. ANSWER ALL QUESTIONS 2. THIS FINAL EXAMINATION IS CONDUCTED VIA ☐ Open book ☒ Closed book 3. STUDENTS ARE **PROHIBITED** TO CONSULT THEIR OWN MATERIAL OR ANY EXTERNAL RESOURCES DURING THE EXAMINATION CONDUCTED VIA CLOSED BOOK

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THIS QUESTION PAPER CONSISTS OF **FIVE (5)** PAGES

Q1 Based on the case study provided in **Figure Q1.1**, answer **Q1(a)-Q1(c).**

A major financial trading platform operates a real-time processing server for handling stock trades, market data updates, and background analytics. The platform's servers handle an immense volume of transactions per second, especially during market opening and closing times when trading activity spikes. Previously, the server's Operating System (OS) used a non-preemptive scheduling approach that processed tasks based on arrival time. This caused high-priority trading tasks to get delayed, thus, lead to missed trade opportunities, frustrating traders who relied on real-time order processing.

The IT team configured the OS to use preemptive scheduling algorithm, which dynamically evaluated each task's remaining execution time despite of their priority. This allowed the OS to handle high-frequency, time-sensitive tasks with shorter durations efficiently. Tasks were categorized as follows:

- High-Priority Tasks: Real-time trade orders, user requests for stock price data, and order cancellations.
- Medium-Priority Tasks: Real-time data updates, like stock price changes and trading volume updates.
- Low-Priority Long Tasks: Background analytics, trend analysis, and scheduled reports.

Figure Q1.1 Case study

(a) Identify **ONE (1)** algorithm that the IT team has employed previously. Justify your answer.

(2 marks)

(b) **Table Q1.1** shows list of tasks with their arrival and execution time. The time quantum is 6s (if required).

Table Q1.1 List of tasks

Tasks	Arrival Time (mm:ss)	Execution Time (s)
Background analytics	00:00	45
Trade order 01	00:02	5
Stock price changes	00:04	2
Trade order 001	00:05	5
Trading volume updates	00:10	8
User request information	00:11	3

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(i) If the average waiting time for all the task listed is 6.5s, identify **ONE (1)** pre-emptive scheduling algorithm that executes the tasks.

(2 marks)

(ii) Based on your answer in **Q1(b)(i)**, illustrate the sequence of all tasks using a Gantt Chart.

(6 marks)

(iii) Assuming that ONE (1) user suddenly cancels his order on 00:09 with 6s execution time.

Calculate the new average waiting time for all the tasks. Draw a Gantt Chart to support your answer.

(6 marks)

(c) Assuming that the trade order task is independent and remains unaffected once it has started until it is completed, identify **TWO (2)** disadvantages that may arise when the new configuration is implemented by the IT team, based on your answers in **Q1(b)(i)-(iii)**.

(4 marks)

Q2 Consider page reference string 7, 5, 2, 4, 5, 7, 9, 6, 5, 7, 5, 2, 8, 6, 2, 5, 7, 5, 2, 6.

(a) Perform the access sequence for the case of a cache capacity of four pages using the following algorithms:

(i) First-In-First-Out (FIFO) page replacement algorithm.

(7 marks)

(ii) Least Recently Used (LRU) page replacement algorithm.

(7 marks)

(b) Based on your answers in **Q2(a)**, calculate the hit rate and the miss rate for the FIFO, and LRU algorithms.

(6 marks)

Q3 Explain the **FOUR (4)** conditions required for deadlock occurrence.

(8 marks)

Q4 Suggest **TWO (2)** deadlock prevention strategies. Which of these is most suitable for a real-time system. Justify your answer.

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(5 marks)

Q5 Illustrate and compare using both concurrent and sequential processing for the following equation: `A = 5 (6+B) + C*D - F`. Your comparison should focus on the number of steps required by each processing method to complete the equation.

(12 marks)

Q6 Given five memory partitions in the following order of `12 MB, 4 MB, 7 MB, 3 MB, and 9 MB`, and a list of active processes as shown in **Table Q6.1**.

Table Q6.1 List of active processes

Process	Memory Required
`Browser`	`2.5 MB`
`Antivirus Update`	`4.5 MB`
`Video Player`	`6 MB`
`Background Service Process`	`1 MB`

(a) Illustrate the new memory partitions after all processes are loaded using the following algorithms:

(i) Best fit.

(5 marks)

(ii) Worst fit.

(5 marks)

(b) Based on your answers in **Q6(a)(i)-(ii)**, calculate the amount of internal fragmentation in each process memory block.

(5 marks)

(c) Illustrate the deallocation of the `Antivirus Update` process with its adjacent free block if the process terminates sooner than the rest based on your answers in **Q6(a)(ii)**.

(5 marks)

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Q7 Based on the given scenario, answer **Q7(a)-Q7(c)**

A company has a shared network drive used by different departments, each requiring different levels of file access. The files are stored using a mix of allocation methods. Employees access files either sequentially or directly, depending on the department's needs.

(a) Discuss the advantages and disadvantages of three methods: contiguous, linked, and indexed.

(6 marks)

(b) Explain how access methods in file systems influence the efficiency of file operations.

(3 marks)

(c) Analyze different directory structures (single-level, and tree-structured) by comparing their suitability in managing large-scale file systems.

(6 marks)

- END OF QUESTIONS -

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